

1. Choose an assignment rule that is random and precise

Three assignment mechanisms

- **Bernoulli:** assign each unit independently,

$$T_i \sim \text{Bernoulli}(\pi).$$

The treated count is random, so a small experiment can be badly unbalanced—or even have no treated units.

- **Complete randomisation:** fix m of N units as treated. There are

$$\binom{N}{m} = \frac{N!}{m!(N-m)!}$$

legal allocations.

- **Blocked randomisation:** form similar groups using pre-treatment covariates, then fix the treated count within each group.

Randomisation identifies effects; restrictions can improve precision.

Allocation affects the standard error

The notes write the standard error of the difference in means as

$$SE(\hat{\tau}) = \sqrt{\frac{1}{N-1} \left[\frac{mS_0^2}{N-m} + \frac{(N-m)S_1^2}{m} + 2S_{01} \right]}.$$

Here S_0^2 and S_1^2 are the finite-population variances of $Y(0)$ and $Y(1)$; S_{01} is their covariance; m and $N-m$ are the treatment and control sample sizes.

- If $S_0^2 = S_1^2$, equal allocation minimises the first two terms.
- If variances differ, place somewhat more units in the higher-variance condition.
- A single random allocation can still align treatment with an important outcome predictor.

Precision is a design problem, not something to repair only after outcomes arrive.

2. Blocking removes predictably bad comparisons

Canvassing and enrolment

The four precincts in the notes have the following potential outcomes:

Precinct	Party	$Y(0)$	$Y(1)$	τ_i
1	Dem	20	60	40
2	Dem	30	70	40
3	Rep	20	30	10
4	Rep	30	40	10
Mean		25	50	25

Thus the true finite-sample effect is

$$\bar{\tau} = \frac{1}{4} \sum_{i=1}^4 [Y_i(1) - Y_i(0)] = 25.$$

We never observe both potential outcomes for a precinct. Assignment determines which treatment–control comparison becomes visible.

Complete randomisation is unbiased but variable

Treating two precincts creates $\binom{4}{2} = 6$ allocations and estimates

40, 35, 25, 25, 15, 10.

Their assignment average is

$$E_T[\hat{\tau}] = \frac{40 + 35 + 25 + 25 + 15 + 10}{6} = 25.$$

The estimator is unbiased, but assignments that treat both Democratic or both Republican precincts confound treatment with party.

Block on party, then randomise within party

Pairing the two Democratic precincts and the two Republican precincts leaves $2^2 = 4$ legal assignments. Their estimates are

35, 25, 25, 15.

The mean stays 25, but the extreme estimates 40 and 10 disappear.

Blocking preserves randomisation while concentrating estimates closer to the truth.

Block only on covariates measured before treatment.

3. Build blocks from strong pre-treatment predictors

What blocking controls

The notes decompose total estimation error into

$$\Delta = \underbrace{\Delta_S}_{\text{sampling error}} + \underbrace{\Delta_T}_{\text{treatment imbalance}},$$

and then into observed and unobserved components:

$$\Delta = \Delta_{SX} + \Delta_{SU} + \Delta_{TX} + \Delta_{TU}.$$

Blocking directly controls treatment imbalance on observed covariates, Δ_{TX} . It may also reduce Δ_{TU} when observed predictors are correlated with omitted predictors.

Why block?

- ▶ better treatment–control covariate balance;
- ▶ estimates closer to the finite-sample truth;
- ▶ greater efficiency when X predicts Y ;
- ▶ block-specific effects and guidance for limited resources.

The notes' design maxim: block, randomise, then adjust.

When exact matching is impossible

Exact blocking works for one or two discrete predictors. For multiple continuous covariates, measure pairwise dissimilarity using

$$MD_{ij} = \sqrt{(x_i - x_j)' \hat{\Sigma}^{-1} (x_i - x_j)}.$$

Mahalanobis distance standardises scale and accounts for correlation among covariates. The workflow is:

1. collect predictors X ;
2. calculate all pairwise distances;
3. create similar pairs or groups;
4. randomise treatment within each block.

Practical safeguards

Use MCD or MVE covariance estimates when outliers distort ordinary scaling. Add calipers or geographic restrictions where distant matches are implausible or nearby units may interfere.

Applied example: Seguro Popular paired 148 health clusters on baseline characteristics before assigning 74 to early roll-out.

More covariates are useful only when they help predict outcomes or define important comparisons.

4. Preserve the assignment mechanism from design through analysis

Sequential experiments need adaptive randomisation

When units arrive over time, compare new unit q with the R_t units already assigned to condition t :

$$MD_{qt} = \frac{1}{R_t} \sum_{r: T_r=t} MD_{qr}.$$

Give q a higher probability of entering the condition to which it is more dissimilar. This improves balance on continuous, correlated covariates while retaining randomness.

A discrete biased-coin rule illustrates the principle:

$$P(T_q = 1 \mid p) = \begin{cases} 2/3, & n_{pT} < n_{pC}, \\ 1/3, & n_{pT} > n_{pC}, \\ 1/2, & n_{pT} = n_{pC}. \end{cases}$$

Treatment is favoured when profile p is underrepresented in treatment, but assignment never becomes deterministic.

The PTSD study in the notes shows how this works under rolling recruitment.

“As ye randomise, so shall ye analyse”

If blocks determined assignment, randomisation inference must preserve them:

1. keep outcomes and the realised blocks fixed;
2. reassign treatment within each block;
3. recompute the effect estimate many times;
4. compare the observed estimate with this null distribution.

Shuffling labels across blocks introduces allocations that the experiment could never have produced.

Allow for unequal assignment probabilities

If block j treats m_j of N_j units, then

$$p_{ij} = \frac{m_j}{N_j}, \quad w_{ij} = \frac{T_i}{p_{ij}} + \frac{1 - T_i}{1 - p_{ij}}.$$

Treated observations receive weight $1/p_{ij}$ and controls receive $1/(1 - p_{ij})$, so blocks with different treatment shares are represented correctly.

Report the actual blocks, assignment probabilities, balance, estimator, and design-consistent uncertainty.